

Supplementary Material

Feature-Constrained Adversarial Robustness of a Hybrid CNN–ECA–Transformer IDS for Encrypted Traffic: A White-Box and Black-Box Evaluation

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S1 Scope of this Supplement

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Table S1: Contents of the supplement and origin of each item.

Section	Content	Origin
S2	Attacker knowledge, query schedule, and shadow-data noise in the two settings	Section IV, Algorithm 2, Table I
S3	Feature-constraint mask: selection criteria and rationale per feature group	Section IV-A
S4	Perturbation budgets in original feature units	Section II (closed form)
S5	Relation between ASR, ASR-I, and FNR; sensitivity to clean recall	Eqs. (7), (12), (13); Tables III–IV
S6	Choice of Welch’s test; 95% confidence intervals of all reported means	Section IV-B; Tables III–IV
S7	Full computational environment	Paper (moved for space)

S2 Threat Model and Black-Box Characterization

Table S2 extends Table II of the paper with the details of the surrogate-based black-box setting given in Section IV-E, Algorithm 2, and Table I.

Table S2: Attacker knowledge and capabilities (extended version of Table II).

Aspect	White-box	Black-box
Weights, gradients of M1	Full access	None
Architecture of M1	Known	Unknown
Decision threshold t^*	Fixed (tuned on validation)	Unknown; fixed
Feature set and Min–Max scaling	Known	Known (shadow data are already normalized)
Feature-constraint mask	Applied to target gradients	Applied to surrogate gradients
Labels used for crafting	True labels y	Surrogate hard pseudo-labels $\mathbb{I}[f_{\text{sub}}(x) \geq 0.5]$
Auxiliary data	Not needed	Random 20% of $\mathcal{D}_{\text{train}}^{\text{bal}}$ of the attacked checkpoint, plus $\mathcal{N}(0, 0.05^2)$ noise, clipped to $[0, 1]$
Oracle output	—	Confidence scores (probabilities)
Query schedule	—	One non-adaptive batch of $Q = X_{\text{shadow}} $ queries per run; no query while crafting
Gradient source	M1	SE–CNN surrogate: $2 \times \text{Conv2D}(64, 128)$, SE block, dense 128, sigmoid; Adam (10^{-3}), 30 epochs, early stopping (patience 5)
Ground-truth test labels	Loss of the attack	Evaluation only

Query schedule. Each run issues exactly one query per shadow flow, i.e., $Q = |X_{\text{shadow}}| = 0.2|\mathcal{D}_{\text{train}}^{\text{bal}}|$, all in a single non-adaptive batch before the attack (Algorithm 2, lines 13–17). Crafting the adversarial examples on the surrogate requires no further query; the resulting examples are then submitted once to M1 for evaluation.

Magnitude of the Gaussian shift. For $\eta \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.05$, the mean absolute change of an unclipped coordinate is $\mathbb{E}|\eta| = \sigma\sqrt{2/\pi} \approx 0.040$, i.e., 4% of the training range, comparable to the largest attack budget ($\epsilon = 0.05$). Clipping to $[0, 1]$ makes the shift one-sided for coordinates at a bound: for a feature at its training minimum (0 after scaling), the clipped noise $\max(0, \eta)$ has mean $\sigma/\sqrt{2\pi} \approx 0.020$ and is never negative. The noise therefore perturbs the auxiliary data in a controlled way; it is not a model of a realistic distribution shift.

Why the setting favors the attacker. (i) The shadow flows are noisy copies of the target’s own training data, so the surrogate is trained in-distribution; (ii) the oracle returns confidence scores rather than labels only; (iii) the attacker knows the feature representation and its normalization. The black-box ASR-I of Table IV therefore characterizes transfer-based evasion under favorable data and oracle assumptions: weaker knowledge (label-only outputs, out-of-distribution auxiliary data, or query limits) would typically reduce transfer, whereas transfer-enhancing attack techniques could increase it. The setting follows the substitute-model paradigm of Papernot *et al.* [1], which also assumes inputs representative of the target’s domain. Surrogate–target agreement and the sensitivity of transfer to σ and to the shadow-set size are not reported (Section VI of the paper).

S3 Feature-Constraint Mask

Criteria. Under the deny-by-default policy of Section IV-A, a feature is perturbable ($m_j = 1$) only if it (i) measures a timing, rate, or size property of the flow and (ii) can plausibly be altered at the source—by delaying, pacing, padding, or fragmenting packets—without violating protocol semantics. Everything else is blocked. Table S3 states, per feature group, the physical meaning, the source-side control mechanism, and the dependencies that the mask does *not* enforce (Section VI of the paper). With these criteria, 30 of the 79 features of HIKARI-2021 and 21 of the 29 features of CIRA-CIC-DoHBrw-2020 are perturbable; the per-feature Boolean mapping of each dataset is released in the code repository.

Table S3: Rationale of the mask per feature group.

Group	Physical meaning	Source-side control	Dependencies (not enforced)	Mask
Timing (e.g., flow duration, inter-arrival times)	Spacing of packets in time	Programmed delays or pacing [2]	Bounded by the flow duration; statistics of the same quantity are linked	Perturbable only if (ii) holds
Rate (e.g., packets or bytes per second)	Volume per unit time	Indirect, through pacing and volume	Ratio of volume (or count) to duration	Perturbable only if (ii) holds
Volume/size (e.g., packet lengths, byte volumes)	Amount of data per packet or flow	Padding or fragmentation [3]	Totals, means, and higher moments of the same quantity are linked; counts are integers	Perturbable only if (ii) holds
Protocol state (e.g., TCP flags, header and window-size fields)	Connection state and stack configuration	Not safely actionable	Governed by the TCP state machine and the peers’ stacks	Blocked
Strong identifiers (IP addresses, ports, timestamps)	Endpoint identity	—	—	Removed before training

S4 Perturbation Budgets in Original Units

Because the Min–Max scaler is fitted on the training split, an ℓ_∞ budget ϵ in normalized space allows each enabled feature j to change by at most

$$\Delta_j(\epsilon) = \epsilon(x_j^{\max} - x_j^{\min}) \quad (\text{S1})$$

in its original unit, for values within the training range $[x_j^{\min}, x_j^{\max}]$. The budgets evaluated in the paper therefore range from 0.1% ($\epsilon = 0.001$) to 5% ($\epsilon = 0.05$) of each feature’s training range. Because Min–Max ranges are set by extreme training values, the same ϵ can be large relative to the typical values of features with extreme outliers and small for features whose range is close to their typical spread.

S5 ASR, ASR-I, and the False-Negative Rate

Notation. Let P be the set of malicious test flows, $\hat{y}^c$ and $\hat{y}^a$ the clean and post-attack predictions, $r = |\{i \in P : \hat{y}_i^c = 1\}|/|P|$ the clean recall, and $\rho = |\{i \in P : \hat{y}_i^c = 0, \hat{y}_i^a = 1\}|/|\{i \in P : \hat{y}_i^c = 0\}|$ the fraction of clean misses that the attack turns into detections. ASR-I is Eq. (12) of the paper, and ASR_P denotes the conventional ASR of Eq. (7) computed over P , as in Fig. 2.

Proposition. $\text{ASR}_P = \text{FNR} = (1-r)(1-\rho) + r \text{ASR-I}$, where FNR is the post-attack false-negative rate (Eq. (13) of the paper).

Proof. A malicious flow is misclassified after the attack either because it was a clean miss that stayed missed, a fraction $(1-\rho)$ of the $(1-r)|P|$ clean misses, or because it was a clean detection turned into a miss, a fraction ASR-I of the $r|P|$ clean detections. Dividing the sum by $|P|$ gives ASR_P , which is by definition the fraction of malicious flows classified as benign after the attack, i.e., the FNR. $\square$

Consequences.

1. Offset: $\text{ASR}_P - \text{ASR-I} = (1-r)(1-\rho - \text{ASR-I})$. At $\varepsilon = 0$ ($\rho = \text{ASR-I} = 0$) it equals the clean miss rate $1-r$ (0.52% for M1 on HIKARI-2021), whereas ASR-I is 0.
2. Sensitivity to clean recall: for fixed ASR-I and $\rho = 0$ the offset grows linearly with $1-r$ (Fig. S1); a detector with $r = 0.8$ and $\text{ASR-I} = 0.3$ reports $\text{ASR}_P = 0.44$.
3. Relation to FNR: $\Delta \text{FNR} = \text{FNR} - (1-r) = r \text{ASR-I} - (1-r)\rho$, so $\text{ASR-I} = \Delta \text{FNR}/r$ when $\rho = 0$: ASR-I is the attack-induced FNR increase normalized by the clean detection rate.
4. Bounds: since $0 \leq \rho \leq 1$, $r \text{ASR-I} \leq \text{FNR} \leq (1-r) + r \text{ASR-I}$.
5. Over all flows, Eq. (7) equals $(\text{FN} + \text{FP})/n$ and also counts benign flows pushed across t^* ; it is therefore not an evasion measure in the IDS sense.

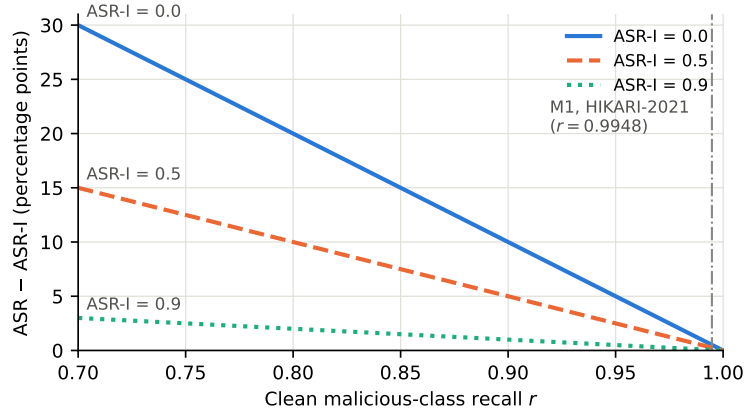


Figure S1: Analytical offset between the conventional ASR over malicious flows and ASR-I as a function of the clean recall r ($\rho = 0$). The dash-dotted line marks M1 on HIKARI-2021 ($r = 0.9948$).

Table S4 applies the bounds of item 4 to HIKARI-2021, the dataset for which the paper reports the clean miss rate ($1-r = 0.52\%$). Values are computed from the run-averaged ASR-I of Tables III–IV; the per-run covariance between r and ASR-I is neglected.

S6 Statistical Analysis

Choice of Welch’s test. The per-run values of the compared conditions have very different variances (e.g., SDs of 0.0001 and 0.045 in Table III), so Student’s equal-variance t -test is inappropriate and Welch’s test was used. The runs are seed-matched—each seed fixes one data split and one trained checkpoint shared by all attacks and budgets—and the unpaired analysis ignores the resulting covariance. When the correlation r_{xy} between paired conditions is positive, the paired standard error $\sqrt{(s_x^2 + s_y^2 - 2r_{xy}s_x s_y)/n}$ is smaller than the unpaired one, so the Welch results are conservative with respect to a paired analysis.

Table S4: Bounds on the post-attack malicious-class FNR (%) for HIKARI-2021 implied by the proposition ($r = 0.9948$; WB = white-box, BB = black-box).

ϵ	WB-FGSM	WB-PGD	BB-FGSM	BB-PGD
0 (clean)	0.52			
0.001	[22.41, 22.93]	[31.40, 31.92]	[12.39, 12.91]	[18.46, 18.98]
0.005	[53.83, 54.35]	[64.89, 65.41]	[33.94, 34.46]	[44.99, 45.51]
0.010	[71.02, 71.54]	[84.01, 84.53]	[56.05, 56.57]	[68.09, 68.61]
0.020	[86.75, 87.27]	[96.01, 96.53]	[73.73, 74.25]	[83.08, 83.60]
0.050	[98.20, 98.72]	[99.02, 99.54]	[89.17, 89.69]	[93.72, 94.24]

Confidence intervals. With $n = 30$ runs, the 95% interval of a reported mean is $\text{mean} \pm t_{0.975,29} \text{SD} / \sqrt{30} = \text{mean} \pm 0.373 \text{SD}$. Tables S5 and S6 list these intervals for every entry of Tables III and IV, computed from the printed means and SDs.

Table S5: 95% confidence intervals of the means in Table III (white-box).

Dataset	ϵ	FGSM F_1	FGSM ASR-I	PGD F_1	PGD ASR-I
HIKARI-2021	0.001	[0.7730, 0.7962]	[0.2175, 0.2331]	[0.7070, 0.7406]	[0.3104, 0.3208]
	0.005	[0.5045, 0.5209]	[0.5280, 0.5542]	[0.3468, 0.3902]	[0.6385, 0.6661]
	0.010	[0.3393, 0.3475]	[0.6982, 0.7296]	[0.1570, 0.2086]	[0.8277, 0.8613]
	0.020	[0.1453, 0.1587]	[0.8541, 0.8899]	[0.0347, 0.0533]	[0.9610, 0.9692]
	0.050	[0.0200, 0.0312]	[0.9852, 0.9890]	[0.00056, 0.00064]	[0.9947, 0.9961]
CIRA-CIC-DoHBrw-2020	0.001	[0.8178, 0.8268]	[0.1804, 0.1916]	[0.7361, 0.7473]	[0.2558, 0.2692]
	0.005	[0.5781, 0.5915]	[0.4386, 0.4550]	[0.3969, 0.4133]	[0.6157, 0.6351]
	0.010	[0.3742, 0.3928]	[0.6715, 0.6931]	[0.2204, 0.2360]	[0.8001, 0.8247]
	0.020	[0.1515, 0.1725]	[0.8532, 0.8764]	[0.0559, 0.0671]	[0.9373, 0.9499]
	0.050	[0.0127, 0.0187]	[0.9685, 0.9775]	[0.00646, 0.00654]	[0.9805, 0.9857]

Clean F_1 (95% CI) — HIKARI-2021: [0.9180, 0.9456]; CIRA-CIC-DoHBrw-2020: [0.9787, 0.9799].

Table S6: 95% confidence intervals of the means in Table IV (black-box).

Dataset	ϵ	FGSM F_1	FGSM ASR-I	PGD F_1	PGD ASR-I
HIKARI-2021	0.001	[0.8386, 0.8700]	[0.1129, 0.1361]	[0.7896, 0.8352]	[0.1766, 0.1946]
	0.005	[0.6932, 0.7312]	[0.3315, 0.3509]	[0.6082, 0.6380]	[0.4377, 0.4669]
	0.010	[0.5196, 0.5666]	[0.5515, 0.5753]	[0.3900, 0.4340]	[0.6714, 0.6976]
	0.020	[0.3556, 0.4086]	[0.7274, 0.7550]	[0.2165, 0.2725]	[0.8198, 0.8504]
	0.050	[0.1607, 0.2093]	[0.8848, 0.9080]	[0.0759, 0.1073]	[0.9328, 0.9514]
CIRA-CIC-DoHBrw-2020	0.001	[0.9095, 0.9155]	[0.0913, 0.0995]	[0.8696, 0.8786]	[0.1369, 0.1481]
	0.005	[0.7786, 0.7898]	[0.2781, 0.2945]	[0.6754, 0.6910]	[0.3857, 0.4051]
	0.010	[0.6048, 0.6220]	[0.4713, 0.4929]	[0.4614, 0.4830]	[0.6001, 0.6247]
	0.020	[0.4205, 0.4437]	[0.6815, 0.7069]	[0.2683, 0.2945]	[0.7798, 0.8074]
	0.050	[0.2012, 0.2296]	[0.8379, 0.8685]	[0.0964, 0.1144]	[0.9132, 0.9296]

S7 Computational Environment

All experiments were conducted on a virtualized server running Ubuntu 24.04.4 LTS (Linux kernel 6.17.0-35-generic), equipped with 24 virtual CPUs based on Intel Xeon Gold 6230 processors at 2.10 GHz, 251.7 GiB of RAM, and a single NVIDIA A2 GPU with approximately 15 GiB of VRAM. The system used NVIDIA driver 535.309.01, with CUDA 12.2 reported by the driver and CUDA Toolkit 12.0 (V12.0.140) available through `nvcc`; primary storage was provided by an approximately 500 GB virtual disk. All times reported in Tables V and VI were measured on this single-GPU configuration, with adversarial examples generated in batches of 64.

References

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